Safety-Critical Cyber-Physical Systems

Autonomous Vehicles | **Mobile Robots** | **Drone Fleets** | **Robot Foundation Models (VLA)**

enables

Thrust 2: Foundation Models for Autonomy

Context-aware perception
Context-Aware Intelligent Vehicles
(GM I-SEE project)
FOCUS: fine-grained driving context

Dependable foundation models
VLA-Scope: shift-aware failure prediction for VLA models
Context Plane for drone fleets, *DroneSys'26*

builds on

Thrust 1: Computing Systems for Robotics

Predictable real-time ML
DNN time variations, *MLBench'23*
Prophet, *RTSS'22* RT-BEV, *RTSS'24*
MM-BEV PP-DNN CPT, *TST'24*
pPerf profiler (NSF CNS-2343601)

Energy-efficient autonomy
E2M, *SEC'19* Donkey, *ICRA'23*
pNav: cyber-physical power management
FEAD fuel prediction, *T-ITS'23*
AyE-Edge, *ICCAD'24* LOTUS, *DAC'24*

Thrust 3: Physics Intelligence

Cross-layer physical safety
When Delay Meets Drift, *CoRL-W'26*

Physical attack plane
Adversarial Calibration Attack on camera–LiDAR fusion

Cyber–physical co-design
Timing, sensing, and dynamics treated as one system

Cross-layer co-design: model · middleware · OS · hardware

Real platforms: Hydra AV · HydraOne · Donkey · Inceptio trucks · Jetson/GPU
Earlier work: SafeShareRide, *SEC'18* · OpenVDAP, *ICDCS'18* · AutoVAPS · Equinox